HOMEWORK 1
46-921

Due Thursday, September 10 at 9:30 AM EDT

Part 1

Revisit the lecture notes, page 15 of Part 1. Row 278303 of the data set appears as extreme in multiple variables. Can you find an explanation for this?

Part 2

In this assignment, you will download some data from the *Wharton Research Data Services (WRDS)* web site and format it. There is a document on the course Canvas page with login details. Read the *WRDS* terms and conditions before using their web site.

While you are logged into the *WRDS* web site, you can look around to see what data are available.

Be sure to submit a Jupyter notebook file with all of your analysis. Also submit the original data file that you downloaded.

After you log in to the *WRDS* site, under the `Get Data` tab choose the `CRSP` database, then `Stock/Security Files`, and then `Monthly Stock File`. Here you will be able to choose what you want to download. For the dates (Step 1), choose September 2000 through June 2015.

For Step 2, choose `TICKER`. This is the ticker symbol. Under "Select an option for entering company codes," type
BBY MDSY
in the `Company Code` box. (This gets the data for Best Buy and ModSys International.)

In Step 3, choose the following itmes in "Identifying Information": `Cusip`, `Company Name`, and `Ticker`. In "Time Series Information," choose `Price`, `Ask or High`, `Bid or Low`, `Closing Bid`, and `Closing Ask`. In "Share Information," choose `Number of Shares Outstanding`. Skip "Delisting Information." In "Distribution Information," choose `Cumulative Factor to Adjust Price`. Skip "NASDAQ Information" and "Market Information."

In Step 4, you choose three formats for the output. Select the `*.csv` format. Don't bother to compress the data. Dates in the form YYMMDDn8. are easy to sort. If you wish, you may enter an email address so that they can notify you when the request has been filled. After you "SUBMIT QUERY", you will probably need to wait several seconds for the response to appear. You may be shown some information indicating that the query is being processed. If you are told that no matches were found, you must have skipped one of the above steps (or added a step that isn't listed).

Once the request is filled, you will be told the name of a data file that you can download.

Use Python to organize the data set as described below. When you submit your assignment, be sure to answer the **boldface** items.

1. The data file will contain a header telling you which columns correspond to which variables. **How many records, i.e. (stock, day) combinations, does the downloaded data file contain, and how many variables (columns) are there?** Make sure that you understand what all of the variables are, even if you have to go back to *WRDS* and look at the variable descriptions.

2. **We requested two different TICKERs. How many distinct TICKER values appeared in the data file?** The ticker symbol for a company's stock can change over time, and the same ticker symbol can be reassigned to a new company after it has been abandoned by another company. A universal unique identifier of stocks is "CUSIP," named for the Committee on Uniform Security Identification Procedures. **How many distinct CUSIP values appeared in the data file?**

3. There are several possible reasons for extra tickers and/or CUSIPs to appear in a data request. Here, we will diagnose the different TICKERs associated with the CUSIP M7037810. **Figure out why the TICKER changed when it did.** You will need to look elsewhere (outside of *WRDS*) to verify your guess. The `COMNAM`, Company Name, might help in this task.

4. The "cumulative factor to adjust price" (`CFACPR`) allows us to make price comparisons over long periods of time. **How many different values of the factor exist in the data file?** If we divide recorded prices `PRC` by `CFACPR`, the prices are adjusted for stock splits and other events. **Before dividing, check to see if there are any records for which the factor is missing (NA) but the price exists.** Compute a new data item (`PRCadj`) for each record by dividing all prices by `CFACPR`, but preserve the original price `PRC` as well. **Give an example of a stock and a day on which an adjustment had to be made (the factor is not 1).**

5. Set all undefined or "not available" adjusted prices to appropriate missing values. **How many "not available" adjusted prices are there?**

6. Determine the set of unique dates in the data set. **How many are there?**

7. Negative price values correspond to certain bits of information being missing, but their absolute values still carry useful price information. **Look up the trading volume data on WRDS for the negative prices in the year 2014 in our data set. What common feature do you notice for the negative prices?** Do a Google search to try to determine how the negative price values can be interpreted. For the remainder of the assignment, convert all negative prices to their absolute values. Do this *both* for `PRC` and `PRCadj`.

8. **Were there any cases in which the Closing Bid (BID) is larger than the Closing Ask (ASK)? If so, offer an explanation for why that happened.**

9. We are interested in January 2001 through June 2015. The extra dates in 2000 were chosen so that we could compute one of the popular momentum measures, namely *3-month return*. For each stock, and each month starting in January 2001, compute the 3-month return (call it `momentum`) as

$$\frac{\text{Adjusted price from previous month}}{\text{Adjusted price from four months earlier}} - 1.$$

   For example, for January 2001, take the ratio of the 31 December 2000 `PRCadj` to the 30 September 2000 `PRCadj`, then subtract 1. This gives the return over the three months that ended on 31 December 2000. We associate this value with January of 2001, because that is when the number becomes available to use.

10. Construct an appropriate plot to compare the distribution of `momentum` between two sets of months: Those where the price increases in the following month versus those where the price decreases in the following month. Construct a separate plot for the two stocks. **Comment on any structure that you see.**

Homework 3

46-921, Fall 2020

Due Thursday, September 24 at 9:30 AM EDT

Please submit a pdf file with your responses to the questions.

Question 1:

Suppose that $X_1, X_2, \ldots, X_n$ are iid with the Gamma(α, β) distribution. Determine the method of moments estimators for α and β.

Comment: This is an important case because maximum likelihood does not admit a closed form for the estimators for α and β.

Question 2:

Assume that $X_1, X_2, \ldots, X_n$ are iid with mean μ and variance σ^2. Both μ and σ^2 are unknown. Assume that the sample mean is $\overline{x} = 14.3$ and the sample standard deviation is $s = 4.2$.

a. Without any further information, can you construct a valid 95% confidence interval for μ if $n = 40$? If so, do it.
b. Without any further information, can you construct a valid 95% confidence interval for μ if $n = 10$? If so, do it.
c. If I told you that the X_i are normal, can you construct a valid 95% confidence interval for μ when $n = 10$? If so, do it.

Question 3:

Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. from the Poisson(λ) distribution.

a. Find the maximum likelihood estimator for λ.
b. Is the estimator found in part (a) an unbiased estimator of λ? Explain.
c. What is the standard error of the estimator found in part (a)?

Question 4:

During lecture we compared three different estimators for λ when working with an iid sample from the Exponential(λ) distribution. We concluded that, based on MSE, the "adjusted" method of moments estimator is the best choice.

Now, conduct a simulation experiment to address the following question: In the case where $n = 20$ and $\lambda = 10$, what proportion of the time does the adjusted method of moments estimator come closer to the true value of λ than does the "worst" of the three estimators (the method of moments estimator based on the second moment)? Be sure to submit your code and results.

Question 5:

Let $X_1, X_2, \ldots, X_n$ be a iid from the following distribution:

$$f_X(x;\theta) = \begin{cases} \dfrac{3x^2}{\theta^3} & 0 \le x \le \theta \\ \\ 0 & \text{otherwise} \end{cases}$$

Here are some facts regarding estimation of θ:

- Method of moments estimator is $\hat{\theta}_1 = \dfrac{4\bar{X}}{3}$
- The maximum likelihood estimator is $\hat{\theta}_2 = X_{(n)}$, where $X_{(n)}$ is the maximum of the X_i.

Here are some useful results about this distribution that will help you do this problem.

- $E(X) = \dfrac{3}{4}\theta$
- $V(X) = \dfrac{3}{80}\theta^2$
- $f_{X_{(n)}}(x) = \dfrac{3nx^{3n-1}}{\theta^{3n}}, \quad 0 \le x \le \theta$
- $V(X_{(n)}) = \dfrac{3n}{(3n+2)(3n+1)^2}\ \theta^2$

Now, do the following:

a. Show that $\hat{\theta}_1$ is an unbiased estimator for θ.

b. Show that $\hat{\theta}_2$ is *not* an unbiased estimator for θ, and find $\text{bias}(\hat{\theta}_2) = [E(\hat{\theta}_2) - \theta]$.

c. Show that for $n > 2$, even though $\hat{\theta}_1$ is unbiased and $\hat{\theta}_2$ is not, $\text{MSE}(\hat{\theta}_2) < \text{MSE}(\hat{\theta}_1)$

46-921, Fall 2020: Homework #4

Due 9:30 AM EDT, Thursday, October 1

Note: When you are asked to approximate the distribution of an estimator, it is **not** sufficient to just say "It is approximately normal."

1. Let $X_1, X_2, \ldots, X_n$ be i.i.d. from the *Pareto distribution*:

$$f_X(x) = \begin{cases} \dfrac{\alpha\lambda^\alpha}{x^{\alpha+1}} & x > \lambda,\ \alpha > 0,\ \lambda > 0 \\ 0 & \text{otherwise} \end{cases}$$

   For this exercise we will assume that λ is a known constant.

   (a) Find the MLE for α, call it $\widehat{\alpha}$.

   (b) Approximate the distribution of $\widehat{\alpha}$.

   (c) A sample of size 100 is taken from a population that we are willing to assume has the Pareto distribution with $\lambda = 2$. It holds that $\sum_i \log(x_i) = 107.39$. Make a statement regarding our best estimate of α and attach a standard error to the esimator.

2. Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. from the Gamma(θ, β) distribution, where θ is unknown and β is known. Even though there is not a closed form for the MLE for θ, one can do the following:

   (a) Derive the Fisher Information $I(\theta)$.

   (b) Approximate the distribution of the MLE for θ.

   (c) Approximate the distribution of the MLE for $\log(\theta)$.

3. Suppose that X is binomial(n, p). The MLE for p is, not surprisingly, the sample proportion $\widehat{p} = X/n$. (You do not need to prove this.)

   (a) What is the MLE for $p/(1-p)$? (This is called the *odds.*)

   (b) Approximate the distribution of the MLE of the odds.

   (c) Use part (b) to construct a $100(1-\alpha)\%$ confidence interval for the odds.

   (d) Write a simulation procedure that tests the validity of the confidence interval found in part (c). Is the confidence interval an adequate approximation when $n = 10$ and $p = 0.10$?

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Homework 5

Due Thursday, October 15 at 9:30 AM EDT

For this homework, turn in both a .ipynb file and a .pdf version.

Question One

This exercise is based on data that appeared in the November 9, 1988 edition of the *Wall Street Journal*. The example was originally used in Siegel (1997), but was also used in Sheather (2009). To quote Siegel:

> US Treasury bonds are among the least risky investments, in terms of the likelihood of your receiving the promised payments. In addition to the primary market auctions by the Treasury, there is an active secondary market in which all outstanding issues can be traded. You would expect to see an increasing relationship between the coupon of the bond, which indicates the size of its periodic payment (twice a year), and the current selling price. The ... data set of coupons and bid prices (are) for US Treasury bonds maturing between 1994 and 1998 ... The bid prices are listed per "face value" of $100 to be paid at maturity. Half of the coupon rate is paid every six months. For example, the first one listed pays $3.50 (half of the 7% coupon rate) every six months until maturity, at which time it pays an additional $100.

The data file can be found on Canvas.

Do each of the following. We will treat "Bid Price" as the response and "Coupon Rate" as the predictor.

1. Use the `seaborn` function `regplot` to create a scatter plot of the bid price versus the coupon rate with the least squares regression line superimposed. Comment on the form.
2. Fit a simple linear regression model to the data using Python.
3. Construct the plot of residuals versus fitted values and comment on its form.
4. Using the standard approach, construct a 95% confidence interval for β_1 based on this initial model. Do you trust this confidence interval?

Question Two

This exercise is based on the "Benchmark Bond Trade Price Challenge" which was conducted on the web site Kaggle.com. You can see the full details of this challenge at

`http://www.kaggle.com/c/benchmark-bond-trade-price-challenge`

but you may have to register (it is free). This is reproduced from the "Background:"

> As far as price transparency is concerned, there has historically been a huge gap between the amount of reference information available to those trading equities versus those trading corporate bonds. Stock exchanges report trades, bids and offers at all times. Free access is available online with a 15 minute delay while traders who demand more

information can pay for ultra efficient real time data and information about size of current bids and offers. By contrast, bond trades are required to be reported within 15 minutes and only those who pay for the TRACE feed can access this information. No quotes are publicly available and the best way to get a quote is to solicit multiple brokers and wait for a reply. Alternatively there are data companies that provide end of day prices, published after the market has closed and with no guarantee that the specific information sought will be included. Accurate bond pricing is also hindered by lack of liquidity. Only a fraction of TRACE eligible bonds trade on a given day, so the most recent trade price is often multiple days old. Pricing bonds based on other more liquid bonds that have similar features is common, but again limited by the presence of such bonds.

Benchmark Solutions is the first provider of realtime corporate bond prices. Every 10 seconds we provide accurate prices that incorporate interest rate data, trades or quotes of the bond in question, trades or quotes of other bonds or CDS of the issuer of the bond in question as well as other input sources. Pricing bonds accurately requires an exacting knowledge of payment schedules, trading calenders and reference data for each bond. This, as well as synthesizing all of the bonds and CDS quotes and trades of a given issuer into implied hazard and funding curves, is something that we feel is beyond the scope of this challenge. Rather, we provide you with a reference price which is an intermediate result of our calculations and is labeled 'curve_based_price' in the dataset. Thus the competition focuses on trading dynamics and microstructure of individual bonds, rather than all bonds from a given issuer.

We created a reduced data set in order to make this challenge something we could manage during this exercise. This data set is availble at

`http://www.stat.cmu.edu/~cschafer/MSCF/bonddata.txt`

We will fit regression models which attempt to model `trade_price`.

1. Read in the data. Make sure to correctly identify and transform any predictors which are categorical. Also, be sure to exclude any variables which are clearly not relevant as predictors in a regression, e.g., ID numbers.

2. Fit a model using all of the predictors. Construct the plot of residuals versus fitted values for this model, and comment on the quality of the fit.

3. Construct the normal probability plot of the residuals. What does this plot tell us?

4. Implement a stepwise variable selection procedure using AIC to find a simpler model. Report on the resulting model.

5. Repeat the above, but use PRESS instead of AIC. How does the resulting model change?